

B5 General Relativity & Cosmology — Model Solutions, 2018

Question 1 — Equivalence principle, four-acceleration, infall onto a black hole

Problem. State the strong equivalence principle. Define four-acceleration via covariant differentiation along a worldline. Show $u^\alpha u_\alpha = -c^2$, $u^\alpha a_\alpha = 0$ and $a^\alpha a_\alpha = g^2$. For radial motion in Schwarzschild, derive the form of a^r and discuss whether Albert can return from $r = \sqrt{GM/g}$.

(a) Strong equivalence principle

In any local freely-falling (inertial) frame, the laws of physics — including gravitation — take their special-relativistic form: gravitation is locally indistinguishable from uniform acceleration, and *all* non-gravitational and gravitational experiments are unaffected by the presence of an external gravitational field.

Locally, all physics is special-relativistic; gravity is a property of spacetime curvature, not a force.

(b) Four-acceleration via covariant derivative

The components V^α live in distinct tangent spaces along the worldline; the coordinate basis ∂_β itself rotates, so $dV^\alpha/d\lambda$ confuses true change with basis change.

The covariant “rate of change along the worldline” is the absolute (covariant) derivative:

$$\frac{DV^\alpha}{D\lambda} = \frac{dV^\alpha}{d\lambda} + \Gamma^\alpha_{\beta\gamma} V^\beta \frac{dx^\gamma}{d\lambda}.$$

Apply with $V^\alpha = u^\alpha$ and $\lambda = \tau$, so $dx^\gamma/d\tau = u^\gamma$:

$$a^\alpha = \frac{Du^\alpha}{D\tau} = \frac{du^\alpha}{d\tau} + \Gamma^\alpha_{\beta\gamma} u^\beta u^\gamma.$$

The Γ ’s are obtained from local first derivatives of the metric.

(c) $u^\alpha u_\alpha = -c^2$, $u^\alpha a_\alpha = 0$, $a^\alpha a_\alpha = g^2$

For a body at rest in its own instantaneous rest frame, $u^\alpha = (c, 0, 0, 0)$ and Minkowski metric $\eta_{\alpha\beta} = \text{diag}(-1, 1, 1, 1)$:

$$u^\alpha u_\alpha = \eta_{\alpha\beta} u^\alpha u^\beta = -c^2.$$

This is a Lorentz scalar, hence frame-independent:

$$u^\alpha u_\alpha = -c^2.$$

Differentiate covariantly along the worldline (metric compatibility, $\nabla g = 0$):

$$\frac{D}{D\tau}(u^\alpha u_\alpha) = 2 u_\alpha \frac{Du^\alpha}{D\tau} = 2 u_\alpha a^\alpha.$$

The l.h.s. is $D(-c^2)/D\tau = 0$:

$$\boxed{u^\alpha a_\alpha = 0.}$$

In Albert's instantaneous rest frame: $u^\alpha = (c, \mathbf{0})$. Orthogonality forces $a^0 = 0$, so $a^\alpha = (0, \mathbf{a})$ with spatial $\mathbf{a}$. He measures proper acceleration $|\mathbf{a}| = g$:

$$a^\alpha a_\alpha = \eta_{ij} a^i a^j = |\mathbf{a}|^2 = g^2.$$

This is again a Lorentz scalar:

$$\boxed{a^\alpha a_\alpha = g^2.}$$

(d) Radial four-acceleration in Schwarzschild

Schwarzschild metric is diagonal; only nonzero Γ 's for purely radial motion ($u^\theta = u^\phi = 0$) are those given. (??) gives the 0- and r -components:

$$a^0 = \frac{du^0}{d\tau} + 2\Gamma^0_{r0} u^r u^0,$$

$$a^r = \frac{du^r}{d\tau} + \Gamma^r_{00} (u^0)^2 + \Gamma^r_{rr} (u^r)^2.$$

Substitute the listed Christoffels with $f \equiv 1 - 2GM/(c^2 r)$, so

$$\Gamma^r_{00} = \frac{GM}{c^2 r^2} f, \quad \Gamma^0_{r0} = -\Gamma^r_{rr} = \frac{GM}{c^2 r^2} f^{-1}.$$

Insert into a^r :

$$a^r = \frac{du^r}{d\tau} + \frac{GM f}{c^2 r^2} (u^0)^2 - \frac{GM f^{-1}}{c^2 r^2} (u^r)^2.$$

Use the normalisation (??) in Schwarzschild, $g_{\alpha\beta} u^\alpha u^\beta = -c^2$, with diagonal metric:

$$-f (u^0)^2 + f^{-1} (u^r)^2 = -c^2.$$

Solve for $(u^0)^2$:

$$(u^0)^2 = \frac{c^2 + f^{-1} (u^r)^2}{f} = \frac{c^2}{f} + \frac{(u^r)^2}{f^2}.$$

Substitute into the a^r expression:

$$a^r = \frac{du^r}{d\tau} + \frac{GM f}{c^2 r^2} \left[\frac{c^2}{f} + \frac{(u^r)^2}{f^2} \right] - \frac{GM f^{-1}}{c^2 r^2} (u^r)^2.$$

Distribute and observe the two $(u^r)^2$ terms cancel:

$$a^r = \frac{du^r}{d\tau} + \frac{GM}{r^2}.$$

$$\boxed{a^r = \frac{du^r}{d\tau} + \frac{GM}{r^2}.$$

The companion equation for a^0 follows analogously:

$$\boxed{a^0 = \frac{du^0}{d\tau} + \frac{2GM}{c^2 r^2} f^{-1} u^r u^0.}$$

(e) Maximum thrust: identifying the equations and Albert's escape

Two scalar conditions fix a^r at maximal acceleration:

(i) magnitude of four-acceleration is g , from (??), with $a^\theta = a^\phi = 0$:

$$a^\alpha a_\alpha = -f (a^0)^2 + f^{-1} (a^r)^2 = g^2;$$

(ii) orthogonality of a^α to u^α , from (??):

$$-f a^0 u^0 + f^{-1} a^r u^r = 0 \implies a^0 = \frac{a^r u^r}{f^2 u^0}.$$

Eliminate a^0 using (ii) in (i):

$$-f \left(\frac{a^r u^r}{f^2 u^0} \right)^2 + f^{-1} (a^r)^2 = g^2.$$

Expand the first term:

$$-\frac{(a^r)^2 (u^r)^2}{f^3 (u^0)^2} + \frac{(a^r)^2}{f} = g^2.$$

Factor $f^{-1} (a^r)^2$:

$$\frac{(a^r)^2}{f} \left[1 - \frac{(u^r)^2}{f^2 (u^0)^2} \right] = g^2.$$

Multiply numerator and denominator inside the bracket by f , and insert $f(u^0)^2 = c^2 + f^{-1} (u^r)^2$ from (d):

$$f^2 (u^0)^2 = f [c^2 + f^{-1} (u^r)^2] = f c^2 + (u^r)^2.$$

Simplify the bracket:

$$1 - \frac{(u^r)^2}{f c^2 + (u^r)^2} = \frac{f c^2}{f c^2 + (u^r)^2}.$$

Substitute back:

$$\frac{(a^r)^2}{f} \cdot \frac{f c^2}{f c^2 + (u^r)^2} = g^2.$$

Cancel f :

$$\frac{(a^r)^2 c^2}{f c^2 + (u^r)^2} = g^2.$$

Solve for $(a^r)^2$:

$$(a^r)^2 = g^2 \frac{f c^2 + (u^r)^2}{c^2} = g^2 \left[f + \frac{(u^r)^2}{c^2} \right].$$

Take the square root:

$$a^r = \pm g \left[\left(1 - \frac{2GM}{c^2 r} \right) + \left(\frac{u^r}{c} \right)^2 \right]^{1/2}.$$

Equations leading to the right-hand side: the four-acceleration norm (??) *plus* the orthogonality (??), combined with the four-velocity normalisation (??).

Can Albert return from $r = \sqrt{GM/g}$? At the turn-around point Albert is momentarily at rest, $u^r = 0$. Combine (??) and (??) (positive root, thrust outwards):

$$\frac{du^r}{d\tau} = -\frac{GM}{r^2} + g \sqrt{1 - \frac{2GM}{c^2 r}}.$$

Outward escape requires the right-hand side to be ≥ 0 at the turning point:

$$g\sqrt{1 - \frac{2GM}{c^2 r}} \geq \frac{GM}{r^2}.$$

Evaluate the Newtonian term at Isaac's radius $r^2 = GM/g$:

$$\frac{GM}{r^2} = \frac{GM}{GM/g} = g.$$

Plug into the escape inequality:

$$g\sqrt{1 - \frac{2GM}{c^2 r}} \geq g.$$

Cancel g and require the square root to be ≥ 1 :

$$\sqrt{1 - \frac{2GM}{c^2 r}} \geq 1.$$

Since $2GM/(c^2 r) > 0$ for finite r , the left-hand side is strictly less than 1:

No — Albert cannot return; he is already past the point of no return.

(Equivalently: the Newtonian-thrust criterion $gr^2 = GM$ marks the static escape limit, but the Schwarzschild factor $\sqrt{1 - 2GM/c^2 r} < 1$ reduces the available outward four-acceleration. Isaac's radius corresponds to equality only in the Newtonian limit; the GR correction makes it just inside the no-return region.)

Question 2 — Cosmic-string-like cylindrical spacetime

Problem. For the line element $ds^2 = -c^2 dt^2 + dr^2 + f^2(r)d\phi^2 + dz^2$, derive the non-zero Christoffels, the Ricci tensor and energy-momentum tensor; show $T^{\alpha\beta} = (c^4 f''/8\pi G f) \text{diag}(-1, 0, 0, 1)$. Use this to obtain the mass per unit length of a cosmic-string-like object with the deficit-angle exterior. Discuss observational signatures.

(a) Christoffels, Ricci, and stress-energy

Metric components:

$$g_{tt} = -c^2, \quad g_{rr} = 1, \quad g_{\phi\phi} = f^2(r), \quad g_{zz} = 1; \quad g^{\phi\phi} = 1/f^2.$$

Determinant:

$$|g| = c^2 f^2, \quad \ln |g| = 2 \ln f + \text{const.}$$

Christoffels from $\Gamma^\alpha_{\beta\gamma} = \frac{1}{2}g^{\alpha\delta}(\partial_\beta g_{\gamma\delta} + \partial_\gamma g_{\beta\delta} - \partial_\delta g_{\beta\gamma})$. Only $g_{\phi\phi}$ has r -dependence.

Compute $\Gamma^r_{\phi\phi}$:

$$\begin{aligned} \Gamma^r_{\phi\phi} &= -\frac{1}{2}g^{rr}\partial_r g_{\phi\phi} = -\frac{1}{2} \cdot 1 \cdot 2ff' \\ \Gamma^r_{\phi\phi} &= -ff'. \end{aligned}$$

Compute $\Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r}$:

$$\begin{aligned} \Gamma^\phi_{r\phi} &= \frac{1}{2}g^{\phi\phi}\partial_r g_{\phi\phi} = \frac{1}{2} \cdot f^{-2} \cdot 2ff' \\ \Gamma^\phi_{r\phi} &= \frac{f'}{f}. \end{aligned}$$

All other Christoffels vanish by inspection (no t, ϕ, z dependence in any g).
Ricci tensor from the formula provided. Only R_{rr} and $R_{\phi\phi}$ are non-zero (given).
Compute R_{rr} . Use:

$$R_{rr} = \frac{1}{2} \partial_r \partial_r \ln |g| - \partial_\gamma \Gamma^\gamma_{rr} + \Gamma^\delta_{r\gamma} \Gamma^\gamma_{r\delta} - \frac{1}{2} \Gamma^\gamma_{rr} \partial_\gamma \ln |g|.$$

$\Gamma^\gamma_{rr} = 0$ for all γ , so middle and last terms drop:

$$R_{rr} = \frac{1}{2} \partial_r^2 (2 \ln f) + \Gamma^\phi_{r\phi} \Gamma^\phi_{r\phi}.$$

Differentiate:

$$\frac{1}{2} \partial_r^2 (2 \ln f) = \partial_r \left(\frac{f'}{f} \right) = \frac{f''}{f} - \left(\frac{f'}{f} \right)^2.$$

Add $(\Gamma^\phi_{r\phi})^2 = (f'/f)^2$:

$$\boxed{R_{rr} = \frac{f''}{f}.$$

Compute $R_{\phi\phi}$. Apply the formula:

$$R_{\phi\phi} = \frac{1}{2} \partial_\phi \partial_\phi \ln |g| - \partial_\gamma \Gamma^\gamma_{\phi\phi} + \Gamma^\delta_{\phi\gamma} \Gamma^\gamma_{\phi\delta} - \frac{1}{2} \Gamma^\gamma_{\phi\phi} \partial_\gamma \ln |g|.$$

First term zero (ϕ -independent). Second:

$$\partial_\gamma \Gamma^\gamma_{\phi\phi} = \partial_r (-f f') = -(f')^2 - f f''.$$

Third (only non-zero term: $\delta = \phi, \gamma = r$ and $\delta = r, \gamma = \phi$):

$$\Gamma^\phi_{\phi r} \Gamma^r_{\phi\phi} + \Gamma^r_{\phi\phi} \Gamma^\phi_{\phi r} = 2 \cdot \frac{f'}{f} \cdot (-f f') = -2(f')^2.$$

Fourth:

$$-\frac{1}{2} \Gamma^r_{\phi\phi} \partial_r \ln |g| = -\frac{1}{2} (-f f') \cdot \frac{2f'}{f} = (f')^2.$$

Sum:

$$R_{\phi\phi} = 0 - [-(f')^2 - f f''] + [-2(f')^2] + (f')^2 = f f''.$$

$$\boxed{R_{\phi\phi} = f f''.$$

Ricci scalar:

$$R = g^{\alpha\beta} R_{\alpha\beta} = g^{rr} R_{rr} + g^{\phi\phi} R_{\phi\phi} = \frac{f''}{f} + \frac{f f''}{f^2} = \frac{2f''}{f}.$$

Einstein tensor $G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R$ (set $\Lambda = 0$):

$$G_{tt} = 0 - \frac{1}{2} (-c^2) \cdot \frac{2f''}{f} = \frac{c^2 f''}{f},$$

$$G_{rr} = \frac{f''}{f} - \frac{1}{2} \cdot 1 \cdot \frac{2f''}{f} = 0,$$

$$G_{\phi\phi} = f f'' - \frac{1}{2} f^2 \cdot \frac{2f''}{f} = 0,$$

$$G_{zz} = 0 - \frac{1}{2} \cdot 1 \cdot \frac{2f''}{f} = -\frac{f''}{f}.$$

Einstein equation $G_{\alpha\beta} = -8\pi G T_{\alpha\beta}/c^4$:

$$T_{\alpha\beta} = -\frac{c^4}{8\pi G} G_{\alpha\beta}.$$

Raise both indices via $T^{\alpha\beta} = g^{\alpha\gamma} g^{\beta\delta} T_{\gamma\delta}$ (diagonal):

$$T^{\alpha\beta} = -\frac{c^4}{8\pi G} G^{\alpha\beta}, \quad G^{\alpha\beta} = g^{\alpha\alpha} g^{\beta\beta} G_{\alpha\beta}.$$

Work in the basis $x^0 = ct$, so $g_{00} = -1$, $g^{00} = -1$, and $G_{00} = G_{tt}/c^2 = f''/f$:

$$G^{00} = (g^{00})^2 G_{00} = \frac{f''}{f}, \quad G^{rr} = 0, \quad G^{\phi\phi} = 0, \quad G^{zz} = -\frac{f''}{f}.$$

Substitute:

$$T^{00} = -\frac{c^4}{8\pi G} \cdot \frac{f''}{f} = -\frac{c^4 f''}{8\pi G f}, \quad T^{zz} = +\frac{c^4 f''}{8\pi G f}.$$

Collect with all four components:

$$T^{\alpha\beta} = \frac{c^4 f''}{8\pi G f} \text{diag}(-1, 0, 0, 1).$$

Physical content: energy density $\rho c^2 = c^4 f''/(8\pi G f)$ and equal-magnitude tension $-p_z = \rho c^2$ along z ; transverse pressures vanish. This is exactly a cosmic-string-like source.

(b) Mass per unit length of a deficit-angle string

Energy per unit length μ is the integral of the energy density over the proper transverse cross-section. Proper area element on a $t, z = \text{const}$ slice is $dA = f(r) dr d\phi$. The energy density (component T^{00} in the orthonormal frame $x^0 = ct$) is $\rho c^2 = c^4 f''/(8\pi G f)$:

$$\mu c^2 = \int_0^{2\pi} d\phi \int_0^R \rho c^2 f(r) dr.$$

Substitute ρc^2 :

$$\mu c^2 = \int_0^{2\pi} d\phi \int_0^R \frac{c^4 f''(r)}{8\pi G f(r)} f(r) dr.$$

Cancel f in the integrand:

$$\mu c^2 = \frac{c^4}{8\pi G} \int_0^{2\pi} d\phi \int_0^R f''(r) dr.$$

Integrate trivially in ϕ and once in r :

$$\mu c^2 = \frac{c^4}{8\pi G} \cdot 2\pi [f'(R) - f'(0)] = \frac{c^4}{4G} [f'(R) - f'(0)].$$

Divide by c^2 :

$$\mu = \frac{c^2}{4G} [f'(R) - f'(0)].$$

Boundary values. The problem states $f'(r) \rightarrow 0$ as $r \rightarrow 0$, so $f'(0) = 0$.

Matching at $r = R$. Both f and f' are assumed continuous at $r = R$ (so that the metric and its first derivatives, hence the Christoffels, are continuous). For $r > R$ the line element gives $f(r) = (1 - \lambda)r$, hence $f'(r) = 1 - \lambda$:

$$f'(R) = 1 - \lambda.$$

Substitute into (??):

$$\mu = \frac{c^2}{4G}(1 - \lambda).$$

(c) Why difficult to detect; observational signatures

Test particles in the exterior $ds^2 = -c^2 dt^2 + dr^2 + (1 - \lambda)^2 r^2 d\phi^2 + dz^2$ feel *no Newtonian gravitational acceleration*: the geodesic equation in (r, ϕ) for static metric coefficients yields $\ddot{r} - (1 - \lambda)^2 r \dot{\phi}^2 = 0$ exactly as for free space with rescaled ϕ . A particle initially at rest stays at rest. Hence the object exerts no local pull, no orbital motion.

The geometry is locally flat but globally conical, with deficit angle $\Delta\phi = 2\pi\lambda$. Detection requires a global observation:

- **Gravitational lensing.** Light rays passing on opposite sides of the string focus, producing double images of background sources separated by an angle $\sim 2\pi\lambda \sim 8\pi G\mu/c^2$, with no magnification asymmetry (unlike a point mass).
- **CMB temperature step (Kaiser–Stebbins effect).** A moving string imprints a sharp linear discontinuity in the CMB temperature, $\Delta T/T \sim 8\pi G\mu v_s/c^3$.
- **Gravitational-wave bursts** from cusps and kinks on cosmic-string loops; characteristic spectrum sought by LIGO/LISA/PTAs.
- **Linear over-densities of galaxies / wakes** in the matter distribution behind a moving string.

Question 3 — Cosmic distance ladder, expansion, fate of the universe

Problem. Describe the cosmic distance ladder. Explain cosmological redshift and how it gives $\dot{R}/R$. Describe a probe of $\ddot{R}$ and explain the geometry/fate connection. For a flat matter-dominated (Einstein–de Sitter) universe, estimate the round-trip light time to a $z = 1$ source and discuss the maximum number of message exchanges. Discuss the same in the real, flat Λ -dominated universe.

(a) Cosmic distance ladder

1. **Astronomical Unit.** Radar ranging to inner planets gives the AU to part-in- 10^9 precision.
2. **Trigonometric parallax.** Earth’s orbit (baseline 2 AU) measures stellar parallaxes to $\sim$ kpc with Hipparcos and to $\sim$ 10 kpc with Gaia. Distance $d = 1 \text{ pc} \times (p/\text{arcsec})^{-1}$.
3. **Main-sequence fitting / spectroscopic parallax.** Calibrate stellar luminosities from nearby parallax stars; place a cluster’s main sequence on the H–R diagram and infer the cluster distance from the apparent–absolute magnitude offset.
4. **Cepheid variables.** Period–luminosity relation $M_V = a \log P + b$, calibrated on Galactic / LMC Cepheids with parallax distances. Reach $\sim$ 30 Mpc with HST.

5. **Tully–Fisher (spirals) and Faber–Jackson / fundamental plane (ellipticals).** Empirical luminosity–rotation/velocity relations calibrated against Cepheids.
6. **Type Ia supernovae.** Standardisable candles ($M_B \approx -19.3$ after light-curve shape correction); reach $z \sim 2$.

Each rung is calibrated against the previous one; errors compound, so the absolute scale of the universe is set by the parallax base.

(b) Cosmological redshift and $\dot{R}/R$

In the FRW line element with co-moving comoving spatial coordinates, photons travel on null geodesics $ds^2 = 0$:

$$c dt = R(t) |d\mathbf{x}|.$$

A wavefront emitted at t_e and received at t_r satisfies $\int_{t_e}^{t_r} dt/R(t) = \text{const}$ along the null geodesic. Consider a successive crest emitted at $t_e + \delta t_e$ and received at $t_r + \delta t_r$:

$$\frac{\delta t_e}{R(t_e)} = \frac{\delta t_r}{R(t_r)}.$$

Periods are related by the same scale-factor ratio:

$$1 + z \equiv \frac{\lambda_r}{\lambda_e} = \frac{\delta t_r}{\delta t_e} = \frac{R(t_r)}{R(t_e)}.$$

$$1 + z = \frac{R(t_0)}{R(t_e)}.$$

For low z , expand $R(t_e) = R(t_0)[1 - H_0(t_0 - t_e) + \dots]$ with $H_0 \equiv \dot{R}/R|_0$. The light travel time is $t_0 - t_e \approx d/c$ for a galaxy at proper distance d :

$$z \approx H_0 d/c \implies cz = H_0 d.$$

Plot redshift versus distance (from the ladder) for many galaxies; the slope is Hubble's constant H_0 , which determines $\dot{R}/R|_0$.

(c) Probing $\ddot{R}$ and the connection to fate

Constraint on $\ddot{R}$ comes from the redshift–distance relation *beyond* linear order. Define the deceleration parameter $q_0 \equiv -\ddot{R}R/\dot{R}^2|_0$. Expand the luminosity distance:

$$d_L(z) = \frac{cz}{H_0} \left[1 + \frac{1}{2}(1 - q_0)z + \mathcal{O}(z^2) \right].$$

Type Ia supernovae out to $z \sim 1$ provide $d_L(z)$, and the deviation from linearity gives q_0 . (1998 high- z SN result: $q_0 < 0$, accelerating expansion.)

Why $\dot{R}/R$ and $\ddot{R}/R$ together fix the future. The Friedmann equation

$$\dot{R}^2 = \frac{8\pi G}{3}\rho R^2 + \frac{1}{3}\Lambda c^2 R^2 - c^2 k$$

gives, at $t = t_0$, with $H_0 = \dot{R}/R$, $\Omega_m = 8\pi G\rho/(3H_0^2)$, $\Omega_\Lambda = \Lambda c^2/(3H_0^2)$, $\Omega_k = -c^2 k/(R_0^2 H_0^2)$:

$$\Omega_m + \Omega_\Lambda + \Omega_k = 1.$$

The acceleration equation

$$\ddot{R}/R = -\frac{4\pi G}{3}(\rho + 3p/c^2) + \frac{1}{3}\Lambda c^2$$

becomes (matter $p = 0$):

$$-q_0 = -\frac{1}{2}\Omega_m + \Omega_\Lambda.$$

Two measurements (H_0 , q_0) plus $\Omega_m + \Omega_\Lambda + \Omega_k = 1$ and equation of state pin all three densities, which determines whether $\dot{R}$ ever vanishes (recollapse) or grows forever. Hence the fate.

(d) Round-trip time in EdS, $z = 1$

Einstein–de Sitter: $k = 0$, $\Lambda = 0$, $\rho \propto R^{-3}$. Friedmann eq $\dot{R}^2 = (8\pi G/3)\rho R^2$ integrates to

$$R(t) \propto t^{2/3}, \quad H = \dot{R}/R = \frac{2}{3t}.$$

Today: $H_0 = 2/(3t_0)$, so $t_0 = 2/(3H_0)$.

The scenario: we send a message now at t_0 ; it reaches them at t_1 ; they reply immediately; their reply reaches us at t_2 . We want the wait $t_2 - t_0$.

The galaxy currently appears at $z = 1$, so light reaching us now was emitted at t_e with $R(t_e)/R(t_0) = 1/(1+z) = 1/2$. With $R \propto t^{2/3}$:

$$t_e = t_0 (1/2)^{3/2} = \frac{t_0}{2\sqrt{2}}.$$

The galaxy's fixed comoving distance from us is set by this past-light null geodesic:

$$\chi = \int_{t_e}^{t_0} \frac{c dt}{R(t)} = \frac{c}{R_0} \int_{t_e}^{t_0} \left(\frac{t_0}{t}\right)^{2/3} dt.$$

Integrate $t^{-2/3}$:

$$\chi = \frac{c t_0^{2/3}}{R_0} \cdot 3[t_0^{1/3} - t_e^{1/3}] = \frac{3c t_0}{R_0} [1 - (t_e/t_0)^{1/3}].$$

Plug $(t_e/t_0)^{1/3} = (2^{-3/2})^{1/3} = 1/\sqrt{2}$:

$$\chi = \frac{3c t_0}{R_0} \left(1 - \frac{1}{\sqrt{2}}\right).$$

Define $\tau \equiv (t/t_0)^{1/3}$; the comoving distance covered between times t_a, t_b ($t_a < t_b$) is $(3ct_0/R_0)(\tau_b - \tau_a)$.

Outgoing leg: $t_0 \rightarrow t_1$. Same χ :

$$\tau_1 - 1 = 1 - \frac{1}{\sqrt{2}} \implies \tau_1 = 2 - \frac{1}{\sqrt{2}} \approx 1.293.$$

Cube:

$$t_1/t_0 \approx 2.161.$$

Return leg: $t_1 \rightarrow t_2$. Same χ again:

$$\tau_2 - \tau_1 = 1 - \frac{1}{\sqrt{2}} \implies \tau_2 = \tau_1 + 1 - \frac{1}{\sqrt{2}}.$$

Substitute τ_1 :

$$\tau_2 = 3 - \frac{2}{\sqrt{2}} = 3 - \sqrt{2} \approx 1.586.$$

Cube:

$$t_2/t_0 \approx 3.99.$$

Round-trip wait:

$$\Delta t_{\text{round}} = t_2 - t_0 \approx 2.99 t_0.$$

Insert $t_0 = 2/(3H_0)$:

$$\Delta t_{\text{round}} \approx 2.99 \times \frac{2}{3H_0} \approx \frac{1.99}{H_0}.$$

With $H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $H_0^{-1} \approx 1.36 \times 10^{10} \text{ yr}$:

$$\Delta t_{\text{round}} \approx 1.99 \times 1.36 \times 10^{10} \text{ yr} \approx 2.7 \times 10^{10} \text{ yr}.$$

Fundamental limit? In EdS the comoving particle horizon is

$$\chi_{\text{ph}}(t) = \frac{3ct}{R(t)} \propto t^{1/3} \rightarrow \infty \quad \text{as } t \rightarrow \infty.$$

So the comoving horizon grows without bound; any comoving observer remains in causal contact forever. After each exchange both civilisations have aged by a finite amount, but in EdS the universe lives forever and signals always arrive eventually.

No fundamental limit: infinitely many exchanges in EdS.

(e) Real Λ -dominated flat universe

Observed universe: $\Omega_m \approx 0.3$, $\Omega_\Lambda \approx 0.7$, flat. Flatness without matter dominance is consistent because $\Omega_m + \Omega_\Lambda = 1$: vacuum energy supplies the missing ρ . The universe is asymptotically de Sitter, $R(t) \propto e^{H_\infty t}$.

Comoving horizon in de Sitter is finite:

$$\chi_{\text{event}} = \int_{t_0}^{\infty} \frac{c dt}{R(t)} = \frac{c}{H_\infty R(t_0)},$$

i.e. a future event horizon. Galaxies at present comoving distance beyond χ_{event} recede faster than light asymptotically; signals exchanged after a finite time can no longer reach them.

Each round trip pushes the partner closer to (or past) the event horizon, since the partner's recession velocity grows exponentially in the de Sitter phase. Eventually one round trip will exceed the remaining proper time before the partner crosses the horizon:

Yes — a finite, computable maximum number of messages, set by the de Sitter event horizon.

Quantitatively: $N_{\text{max}} \sim \log_2(H_\infty t_0)/\log_2(1 + z_e)$ where z_e is the recession redshift each round trip; the count is finite even though the individual delays diverge.

Question 4 — Cosmic neutrino background and neutrino mass bound

Problem. Argue $\mu_i = 0$ for relativistic species in the early universe. Compute relativistic N_i, U_i, P_i and entropy. Show electron-to-photon entropy ratio is 7/8. Explain e^+e^- annihilation reheats photons by a known factor. Estimate T_ν today, n_ν today, and use $\sum m_\nu < \rho_c/n_\nu$ to bound the neutrino masses.

(a) Why $\mu_i = 0$ for relativistic species

Photons. Photon number is not conserved in equilibrium reactions like $e^+e^- \leftrightarrow 2\gamma$. Equilibrium of any reaction creating or destroying species i enforces $\sum \mu_i = 0$ over the reactants = products; since photons can be created/destroyed singly in scattering off charges with no chemical potential offset, $\mu_\gamma = 0$.

Other relativistic species at $T = 5 \times 10^9$ K ($k_B T \approx 0.4$ MeV): electrons, positrons, three flavours of neutrinos and antineutrinos. The relevant test is $T \gg m_i c^2 / k_B$: $m_e c^2 = 0.511$ MeV so electrons/positrons are mildly relativistic; neutrinos with $m \lesssim$ eV are ultra-relativistic.

For each particle–antiparticle pair, equilibrium under pair annihilation $i\bar{i} \leftrightarrow \gamma\gamma$ enforces $\mu_i + \mu_{\bar{i}} = 2\mu_\gamma = 0$. Charge/lepton-number near-symmetry of the universe means $n_i \approx n_{\bar{i}}$, hence $\mu_i \approx -\mu_{\bar{i}} \approx 0$ to leading order:

$$\mu_i \approx 0 \text{ for all relativistic species in equilibrium with the photon bath.}$$

(b) Number, energy, pressure for relativistic species

For $\mu_i = 0$, $\epsilon = pc$, integrate over angle $\int d\Omega = 4\pi$:

$$N_i = \int f_i d^3x d^3p = V \frac{g_i}{(2\pi\hbar)^3} 4\pi \int_0^\infty \frac{p^2 dp}{e^{pc/k_B T} \pm 1}.$$

Change variable $x = pc/(k_B T)$:

$$N_i = V \frac{g_i}{(2\pi\hbar)^3} \cdot 4\pi \left(\frac{k_B T}{c} \right)^3 \int_0^\infty \frac{x^2 dx}{e^x \pm 1}.$$

Evaluate the dimensionless integrals:

$$\int_0^\infty \frac{x^2 dx}{e^x - 1} = 2\zeta(3) \approx 2.404 \quad (\text{boson}),$$

$$\int_0^\infty \frac{x^2 dx}{e^x + 1} = \frac{3}{4} \cdot 2\zeta(3) = \frac{3}{2}\zeta(3) \quad (\text{fermion, using the hint}).$$

Simplify the prefactor $g_i \cdot 4\pi/(2\pi\hbar)^3 = g_i/(2\pi^2\hbar^3)$:

$$N_i = V \frac{g_i}{2\pi^2} \left(\frac{k_B T}{\hbar c} \right)^3 \int_0^\infty \frac{x^2 dx}{e^x \pm 1}.$$

Collect:

$$\frac{N_i}{V} = \frac{g_i}{\pi^2} \zeta(3) \left(\frac{k_B T}{\hbar c} \right)^3 \times \begin{cases} 1 & \text{boson} \\ 3/4 & \text{fermion} \end{cases}.$$

Energy density (insert $\epsilon = pc$):

$$\frac{U_i}{V} = \frac{g_i}{(2\pi\hbar)^3} 4\pi \int_0^\infty \frac{p^3 c dp}{e^{pc/k_B T} \pm 1} = \frac{g_i}{2\pi^2} \frac{(k_B T)^4}{(\hbar c)^3} \int_0^\infty \frac{x^3 dx}{e^x \pm 1}.$$

Evaluate (boson: $\pi^4/15$; fermion: $\frac{7}{8} \cdot \pi^4/15$):

$$\frac{U_i}{V} = \frac{\pi^2}{30} g_i \frac{(k_B T)^4}{(\hbar c)^3} \times \begin{cases} 1 \\ 7/8 \end{cases}.$$

Pressure for an ultra-relativistic gas:

$$P_i = \frac{1}{3} \frac{U_i}{V}.$$

(c) Entropy and electron/photon ratio

For $\mu_i = 0$ (per the question's formula):

$$S_i = \frac{1}{T}(U_i + P_i V) = \frac{U_i}{T} \left(1 + \frac{1}{3}\right) = \frac{4U_i}{3T}.$$

Use (??):

$$\boxed{\frac{S_i}{V} = \frac{2\pi^2}{45} g_i k_B \left(\frac{k_B T}{\hbar c}\right)^3 \times \begin{cases} 1 \\ 7/8 \end{cases} .}$$

Hence $S_i \propto T^3$ at fixed comoving volume.

Electron/photon entropy ratio. Photons: $g_\gamma = 2$, boson. Electron+positron: $g_{e^\pm} = 2 + 2 = 4$, fermion. Take the ratio of (??):

$$\frac{S_{e^\pm}}{S_\gamma} = \frac{4 \cdot (7/8)}{2 \cdot 1} = \frac{4 \cdot 7}{2 \cdot 8} = \frac{7}{4}.$$

The question states “ratio of the entropy of the electrons to that of the photons” — if “electrons” means *electrons only* (not positrons), $g_e = 2$:

$$\boxed{\frac{S_e}{S_\gamma} = \frac{2 \cdot (7/8)}{2 \cdot 1} = \frac{7}{8} .}$$

(d) Photon temperature rise after e^+e^- annihilation

When $k_B T$ drops below $m_e c^2 = 0.511 \text{ MeV}$, e^+e^- pairs annihilate into photons. By that epoch neutrinos have decoupled (decoupling $T_\nu^{\text{dec}} \approx 1 \text{ MeV}$, before e^+e^- annihilation): the entropy released into photons is *not* shared with neutrinos.

Entropy conservation in a comoving volume separately for the $\gamma + e^\pm$ sector and the ν sector:

$$S_\gamma^{\text{before}} + S_{e^\pm}^{\text{before}} = S_\gamma^{\text{after}}.$$

Use (??) with $g_\gamma = 2$ (boson), $g_{e^\pm} = 4$ (fermion). Before annihilation $T_\gamma = T_e \equiv T_-$; after, $T_\gamma \equiv T_+$:

$$(2 \cdot 1 + 4 \cdot \frac{7}{8}) T_-^3 = 2 T_+^3.$$

Simplify:

$$(2 + \frac{7}{2}) T_-^3 = 2 T_+^3 \implies \frac{11}{2} T_-^3 = 2 T_+^3.$$

Solve:

$$\boxed{\frac{T_+}{T_-} = \left(\frac{11}{4}\right)^{1/3} \approx 1.401.}$$

Neutrinos — already decoupled — continue redshifting as $T_\nu \propto R^{-1}$, with $T_\nu = T_-$ at the moment of decoupling but no benefit from the entropy dump.

(e) **Today's T_ν , n_ν , neutrino-mass bound**

Apply (??): from the decoupling moment onward, both T_γ and T_ν redshift as $1/R$, but $T_\nu/T_\gamma = (4/11)^{1/3}$ is preserved forever after. With $T_{\gamma,0} = T_{\text{CMB}} = 2.725 \text{ K}$:

$$T_{\nu,0} = T_{\gamma,0} \left(\frac{4}{11} \right)^{1/3}.$$

Numerical:

$$(4/11)^{1/3} = 0.7138.$$

$$T_{\nu,0} \approx 0.7138 \times 2.725 \text{ K} \approx 1.95 \text{ K}.$$

Number density per neutrino species (today). Apply the fermion form of (??). Per flavour, count one helicity of ν and one of $\bar{\nu}$, giving $g_{\nu+\bar{\nu}} = 2$:

$$n_\nu^{\text{flavour}} = \frac{g_{\nu+\bar{\nu}} \zeta(3)}{\pi^2} \cdot \frac{3}{4} \left(\frac{k_B T_{\nu,0}}{\hbar c} \right)^3 = \frac{3\zeta(3)}{2\pi^2} \left(\frac{k_B T_{\nu,0}}{\hbar c} \right)^3.$$

Compute $k_B T_{\nu,0}$:

$$k_B T_{\nu,0} = 1.381 \times 10^{-23} \text{ J K}^{-1} \times 1.95 \text{ K} = 2.69 \times 10^{-23} \text{ J}.$$

Recall $\hbar c$:

$$\hbar c = 3.16 \times 10^{-26} \text{ J m}.$$

Divide:

$$\frac{k_B T_{\nu,0}}{\hbar c} = \frac{2.69 \times 10^{-23}}{3.16 \times 10^{-26}} \text{ m}^{-1} = 8.51 \times 10^2 \text{ m}^{-1}.$$

Cube:

$$\left(\frac{k_B T_{\nu,0}}{\hbar c} \right)^3 = 6.16 \times 10^8 \text{ m}^{-3}.$$

Compute the prefactor:

$$\frac{3\zeta(3)}{2\pi^2} = \frac{3 \times 1.202}{2\pi^2} = \frac{3.606}{19.74} = 0.1827.$$

Multiply:

$$n_\nu^{\text{flavour}} = 0.1827 \times 6.16 \times 10^8 \text{ m}^{-3} \approx 1.12 \times 10^8 \text{ m}^{-3} \approx 112 \text{ cm}^{-3}.$$

Three flavours:

$$n_\nu^{\text{tot}} \approx 3 \times 112 \text{ cm}^{-3} \approx 336 \text{ cm}^{-3}.$$

(Standard textbook: 339 cm^{-3} — agreement.)

Neutrino-mass bound. If neutrinos are massive but non-relativistic today, their contribution to the present density is $\rho_\nu c^2 = \sum_i m_i c^2 n_{\nu,i}$ where $n_{\nu,i} \approx 112 \text{ cm}^{-3}$ per flavour (number density was set when relativistic and is preserved by free-streaming).

Critical density:

$$\rho_c = \frac{3H_0^2}{8\pi G}.$$

$$H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1} = 72 \times 10^3 \text{ m s}^{-1} / 3.086 \times 10^{22} \text{ m} = 2.33 \times 10^{-18} \text{ s}^{-1};$$

$$H_0^2 = 5.43 \times 10^{-36} \text{ s}^{-2}.$$

$$\rho_c = \frac{3 \cdot 5.43 \times 10^{-36}}{8\pi \cdot 6.67 \times 10^{-11}} = \frac{1.63 \times 10^{-35}}{1.676 \times 10^{-9}} \text{ kg m}^{-3}.$$

$$\rho_c \approx 9.74 \times 10^{-27} \text{ kg m}^{-3}.$$

Bound (require $\Omega_\nu \leq 1$, i.e. neutrinos do not exceed the critical density):

$$\sum_i m_i n_{\nu,i} \leq \rho_c.$$

With identical $n_{\nu,i} \equiv n_\nu^{\text{flavour}} \approx 1.12 \times 10^8 \text{ m}^{-3}$:

$$\sum_i m_i \leq \frac{\rho_c}{n_\nu^{\text{flavour}}} = \frac{9.74 \times 10^{-27} \text{ kg m}^{-3}}{1.12 \times 10^8 \text{ m}^{-3}}.$$

Evaluate:

$$\sum_i m_i \leq 8.70 \times 10^{-35} \text{ kg}.$$

Convert to eV/c^2 using $1 \text{ eV}/c^2 = 1.783 \times 10^{-36} \text{ kg}$:

$$\sum_i m_i c^2 \leq \frac{8.70 \times 10^{-35} \text{ kg}}{1.783 \times 10^{-36} \text{ kg}/(\text{eV } c^{-2})} \text{ eV}.$$

$$\boxed{\sum_i m_{\nu_i} c^2 \lesssim 49 \text{ eV}.}$$

(Cosmological data — CMB+LSS — tighten this to $\sum m_\nu \lesssim 0.12 \text{ eV}$, but the simple critical-density argument already excludes any single species above tens of eV.)